

B Complex Instruction Formats in TOD-ProcBench

In this section, we present a symbolic and natural language example of each of the three complex instruction formats, f_1 (Figure 4), f_2 (Figure 5), f_3 (Figure 6), in TOD-ProcBench.

Let C be the set of all conditions and A be the set of all actions. For a nested structure with depth d , each flattened condition-action pair is defined as:

$$(C_n, \mathbf{a}_n) \in \{(c_1 \wedge \dots \wedge c_k, \langle a_1, \dots, a_m \rangle) | c_j \in C, a_\ell \in A\}$$

where $C_n = c_1 \wedge \dots \wedge c_k$ represents a conjunction of composite conditions, and $\mathbf{a}_n = \langle a_1, \dots, a_m \rangle$ represents a sequence of actions that must be executed when C_n is satisfied.

For example, the nested structure in f_1 [escapeinside=(**)] If (* c_1 *) : (* a_1 *) If (* c_2 *) : (* a_3 *) (* a_4 *) (* a_2 *) If (* c_3 *) : (* a_5 *)

is flattened in f_2 to:

$$\{(c_1, \langle a_1 \rangle), (c_1 \wedge c_2, \langle a_3, a_4 \rangle), (c_1, \langle a_2 \rangle), (c_3, \langle a_5 \rangle)\}$$

.

A flattened structure in f_2 $\{(c_1, \langle a_1, a_2 \rangle)\}$ is then transformed in f_3 to $\{\text{"conditions"} : [c_1], \text{"actions"} : [a_1, a_2]\}$.

```
<What_to_do>
If customer received an email confirmation showing incorrect quantity
of items ordered:
  - Identify customer by asking for full name or account ID
  - Get order ID from customer
  - Verify customer's identity
  - Check system records to confirm actual quantity ordered
  - If email was incorrect:
    - Inform customer of the correct quantity
      ordered based on system records
    - If order has already shipped:
      - Check shipping status
      - If delivered, start return process for extra item(s)
      - If not delivered yet, inform customer to contact again after
        receiving to initiate return
    - Offer refund for extra item(s) charged
  - If email was correct:
    - Explain reason for extra item(s) if possible
    - Start return process if applicable
</What_to_do>
```

Figure 4: Example of a TOD-ProcBench instruction document in format f_1 .

```

<What_to_do>
If customer received an email confirmation showing incorrect quantity
of items ordered:
  - Identify customer by asking for full name or account ID
  - Get order ID from customer
  - Verify customer's identity
  - Check system records to confirm actual quantity ordered

If customer received an email confirmation showing incorrect quantity
of items ordered
AND email was incorrect:
  - Inform customer of the correct quantity ordered based on
    system records

If customer received an email confirmation showing incorrect quantity
of items ordered
AND email was incorrect AND order has already shipped:
  - Check shipping status

If customer received an email confirmation showing incorrect quantity
of items ordered
AND email was incorrect AND order has already shipped AND delivered:
  - Start return process for extra item(s)

If customer received an email confirmation showing incorrect quantity
of items ordered
AND email was incorrect AND order has already shipped AND not delivered yet:
  - Inform customer to contact again after receiving to initiate return

If customer received an email confirmation showing incorrect quantity
of items ordered
AND email was incorrect:
  - Offer refund for extra item(s) charged

If customer received an email confirmation showing incorrect quantity
of items ordered
AND email was correct:
  - Explain reason for extra item(s) if possible
  - Start return process if applicable
</What_to_do>

```

Figure 5: Example of a TOD-ProcBench instruction document in format f_2 .

```

<What_to_do>
[
  {
    "Conditions": [
      "customer received an email confirmation showing incorrect quantity of
        items ordered"
    ],
    "Actions": [
      "Identify customer by asking for full name or account ID",
      "Get order ID from customer",
      "Verify customer's identity",
      "Check system records to confirm actual quantity ordered"
    ]
  },
  {
    "Conditions": [
      "customer received an email confirmation showing incorrect quantity of
        items ordered",
      "email was incorrect"
    ],
    "Actions": [
      "Inform customer of the correct quantity ordered based on system records"
    ]
  },
  {
    "Conditions": [
      "customer received an email confirmation showing incorrect quantity of
        items ordered",
      "email was incorrect",
      "order has already shipped"
    ],
    "Actions": [
      "Check shipping status"
    ]
  },
  {
    "Conditions": [
      "customer received an email confirmation showing incorrect quantity of
        items ordered",
      "email was incorrect",
      "order has already shipped",
      "delivered"
    ],
    "Actions": [
      "Start return process for extra item(s)"
    ]
  },
  {
    "Conditions": [
      "customer received an email confirmation showing incorrect quantity of
        items ordered",
      "email was incorrect",
      "order has already shipped",
      "not delivered yet"
    ],
    "Actions": [
      "Inform customer to contact again after receiving to initiate return"
    ]
  },
  {
    "Conditions": [
      "customer received an email confirmation showing incorrect quantity of
        items ordered",
      "email was incorrect"
    ],
    "Actions": [
      "Offer refund for extra item(s) charged"
    ]
  },
  {
    "Conditions": [
      "customer received an email confirmation showing incorrect quantity of
        items ordered",
      "email was correct"
    ],
    "Actions": [
      "Explain reason for extra item(s) if possible",
      "Start return process if applicable"
    ]
  }
]
</What_to_do>

```

Figure 6: Example of a TOD-ProcBench instruction document in format f_3 .